

CURRICULUM VITAE: KYLE GWIRTZ

Assistant Research Scientist UMBC/NASA GSFC
NASA Goddard Space Flight Center
Greenbelt, Maryland

Tel. 785-341-2933
kyle.gwirtz@nasa.gov
<https://kjg136.github.io>

EXPERIENCE

Research experience in data assimilation, machine learning and inverse problems. Specific focus on developing models and methods for geomagnetic data assimilation, forecasting and the investigation of geomagnetic reversals and excursions.

EDUCATION

Scripps Institution of Oceanography	Earth Sciences	Ph.D.	2021
University of Arizona	Applied Mathematics	M.S.	2019
University of Kansas	Mathematics	M.A.	2012
University of Kansas	Mathematics	B.Sc.	2009

POSITIONS

Assistant Research Scientist, UMBC/NASA GSFC	2024-present
Postdoctoral Researcher, NASA Goddard Space Flight Center	2022-2024
Graduate Research Assistant, Scripps Institution of Oceanography	2020-2021
Graduate Research Assistant, University of Arizona	2018-2020
Graduate Intern, NASA Goddard Space Flight Center	Summer 2018
Graduate Teaching Assistant, University of Arizona	2016-2018

AWARDS

NASA ROSES ESI Grant 2023 (Science PI)	2024-present
NASA Postdoctoral Fellowship (NPP)	2022-2024
Student Author Award, Geophysical Journal International	2021
NASA Earth and Space Science Fellowship (NESSF)	2018-2021

PROFESSIONAL ACTIVITIES

Session Co-Chair: SWARM 10th anniversary (Core Dynamics)	Spring 2024
Session organizer: SIAM-Mathematics of Planet Earth	Summer 2020
Reviewer: Physics of the Earth and Planetary Interiors, IUGG special publications, Springer Nature, Geophysical Research Letters, Journal of Geophysical Research, Earth, Planets and Space	

PUBLICATIONS

K. Gwirtz, W. Kuang, C. Yi, A. Tangborn, *Impact of localization and inflation on geomagnetic data assimilation*, Physics of the Earth and Planetary Interiors, 355, (2024)

W. Kuang, K. Gwirtz, A. Tangborn, M. Morzfeld *Understanding and predicting geomagnetic*

secular variation via data assimilation, IUGG special publication series, “Applications of data assimilation and inverse problems in the Earth Sciences”, chap 4.1, editors A. Ismail-Zadeh, F. Castelli, D. Jones, S. Sanchez, (2023)

K. Gwirtz, T. Davis, M. Morzfeld, C. Constable, A. Fournier, G. Hulot, *Can machine learning reveal precursors of reversals of the geomagnetic axial dipole field?*, Geophysical Journal International, 231(1), 520-535 (2022).

K. Gwirtz, M. Morzfeld, W. Kuang, A. Tangborn, *A testbed for geomagnetic data assimilation*, Geophysical Journal International, 227(3), 2180-2203 (2021).

K. Gwirtz, M. Morzfeld, A. Fournier, G. Hulot, *Can one use Earth’s magnetic axial dipole field intensity to predict reversals?*, Geophysical Journal International, 225(1), 277-297 (2021).

M. Brio, J.G. Caputo, K. Gwirtz, J. Liu and A. Maimistov, *Scattering of a short electromagnetic pulse from a Lorentz-Duffing film: theoretical and numerical analysis*, Wave Motion, 89, 43-56 (2019).

SELECTED PRESENTATIONS

European Geosciences Union General Assembly	Spring 2024
Title of Poster: Estimating core dynamics via the assimilation of magnetic field models into numerical dynamos	

SWARM 10th Anniversary Meeting	Spring 2024
Title of Poster: Estimating core dynamics via the assimilation of magnetic field models into numerical dynamos	

American Geophysical Union Fall Meeting	Fall 2023
Title of Poster: Studying the Earth’s core via the assimilation of geomagnetic observations into numerical dynamo models	

International Union of Geodesy and Geophysics General Assembly	Summer 2023
Title of Talk: Localization and inflation in geomagnetic data assimilation: estimating the dynamic state of the Earth’s outer core	

NASA Solid-Earth Team Meeting	Fall 2022
Title of Poster: Geomagnetic data assimilation	

NASA GSFC 61A seminar	Fall 2022
Title of Talk: Data assimilation and machine learning in the Earth Sciences: lessons from applications to geomagnetism	

SEDI Symposium	Summer 2022
Title of Poster: Searching for precursors of geomagnetic reversals with machine learning	

American Geophysical Union Fall 2021 Meeting	Fall 2021
Title of Poster: Predicting Geomagnetic Reversals with Machine Learning	
Title of Poster: A testbed for geomagnetic data assimilation	

Joint ICTP-IUGG Workshop on Data Assimilation and Inverse Problems	Fall 2021
Title of Talk: Understanding and predicting geomagnetic secular variation via data assimilation	

SIAM-Computational Science and Engineering	Spring 2021
Title of Talk: Geomagnetic data assimilation for decadal scale forecasts: lessons from a new	

simplified model

Data Driven Discovery Showcase, University of Arizona	Spring 2021
Title of Talk: Investigating the predictability of Earth's magnetic field	
American Geophysical Union Fall 2020 Meeting	Fall 2020
Title of Poster: Intensity based predictions of the dipole field and their value in characterizing the Earth-like nature of models	
SEDI Symposium	Fall 2020
Title of Poster: Can one use Earth's magnetic axial dipole field intensity to predict reversals?	
SIAM-Mathematics of Planet Earth	Summer 2020
Title of Talk: Data assimilation experiments with a reduced-order model of the geodynamo	
Scripps Institution of Oceanography, Paleomagnetism Seminar	Summer 2020
Title of Talk: Can one use Earth's magnetic axial dipole field intensity to predict reversals?	
American Geophysical Union Fall 2019 Meeting	Fall 2019
Title of Poster: A reduced-scale model for understanding the numerics of geomagnetic data assimilation	
NASA Solid-Earth Team Meeting	Fall 2019
Title of Poster: Geomagnetic data assimilation: numerical experiments with a reduced-scale model	
International Union of Geodesy and Geophysics General Assembly	Summer 2019
Title of Poster: Geomagnetic data assimilation: numerical experiments with a reduced-scale model	
American Geophysical Union Fall 2018 Meeting	Fall 2018
Title of Poster: Localization and bias correction in geomagnetic data assimilation: systematic numerical experiments with reduced-scale models	
U2 can UQ, University of Arizona	Spring 2018
Title of Talk: Geomagnetic data assimilation	

SELECTED TEACHING

Graduate Teaching Assistant, Department of Mathematics, University of Arizona

<i>Semester</i>	<i>Course</i>
Spring 2018	Math 120: Precalculus
Fall 2017	Math 263: Introduction to Statistics and Biostatistics
Spring 2017	Math 112: College Algebra
Fall 2016	Math 112: College Algebra

Graduate Teaching Assistant, Department of Mathematics, University of Kansas

<i>Semester</i>	<i>Course</i>
Spring 2012	Math 115: Calculus I
Fall 2011	Math 116: Calculus II
Summer 2011	Math 115: Calculus I
Spring 2011	Math 104: Precalculus
Fall 2010	Math 104: Precalculus
Spring 2010	Math 101: College Algebra
Fall 2009	Math 115: Calculus I